Fiscal Year:	FY 2021	**Task Last Updated:**	FY 02/15/2021
PI Name:	Davis, Catherine M. Ph.D.		
Project Title:	VNSCOR: Mechanisms of Radiation-Induced Changes in Sustained Attention and Social Processing		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Organization Name:	Henry M. Jackson Foundation		
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Zip Code:	20817-1891	**Congressional District:**	8
Comments:	Campus address (Jan 2022): Department of Pharmacology and Molecular Therapeutics, Uniformed Services University of the Health Sciences, 4301 Jones Bridge Road, Bethesda, MD 20814. NOTE: PI formerly at Johns Hopkins University; moved to Henry M. Jackson Foundation for the Advancement of Military Medicine in fall 2020.		
Project Type:	Ground	**Solicitation / Funding Source:**	2016-2017 HERO NNJ16ZSA001N-SRHHC. Appendix E: Space Radiobiology and Human Health Countermeasures Topics
Start Date:	04/15/2018	**End Date:**	08/31/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 08/31/2025 per L. Juliette/JSC (Ed., 4/29/22) NOTE: End date changed to 10/1/2020 (officially 9/30/2020) due to PI move to Henry M. Jackson Foundation in fall 2020 (Ed., 1/18/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hienz, Robert Ph.D. (Johns Hopkins University) Robinson, Siobhan Ph.D. (Hamilton College)		
Grant/Contract No.:	80NSSC18K1080		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	[Ed. note Jan 2019: See also project, "VNSCOR: Responses of the Nervous System to Chronic, Low Dose Charged Particle Irradiation" (Principal Investigator (PI): Greg Nelson)] [Ed. note Jan 2022: Project continues with the same PI, Catherine Davis, with grant number 80NSSC22K0022, due to PI move to Henry M. Jackson Foundation.] NELSON/DAVIS VIRTUAL NASA Specialized Center of Research (NSCOR): The project is organized as 5 large experimental campaigns to quantify responses for an interrelated set of central nervous system (CNS) outcome measures in mice to acute and protracted exposures to protons at a dose of 0.5 Gy and sham controls; acute and protracted exposures to 0.25 and 0.5 Gy of charged particles; and acute and protracted exposures to 0.5 and 1.5 Gy of gamma rays. This research builds on previous studies that demonstrated that proton and HZE (high charge energy) exposures result in individual differences in deficits in sustained attention, but more general deficits in recognition memory. This current project is combined with "Responses to the Nervous System to Chronic, Low Dosed Charged Particle Irradiation" (PI: Nelson) in order to explore if these effects are LET (linear energy transfer)-dependent for 16O ions, add a relatively understudied, but important, ion (4He), and examine CNS effects in whole animals following fractionated exposures, and the interaction of other space flight factors (e.g., sleep fragmentation). Revised Specific Aims: Aim 1a: Effects of protracted exposure to five-ion GCR [galactic cosmic rays] sim (1H, 4He, 28Si, 16O, 56Fe) Aim 1b: Effects of protracted exposure to GCR sim without protons (4He, 28Si, 16O, 56Fe) Aim 1c: Effects of protracted exposure to protons only Aim 1d: Acute exposure to 4He (250 MeV/n) Aim 2a: Neuronal activation and molecular markers following radiation Aim 2b: Chemogenetic silencing of mPFC subregions Specific Aims: 1) Determine the effects of acute, single 16O and 4He ion exposures on sustained attention, social odor recognition memory, and social dominance. (This aim has been modified in order to integrate with Nelson project); 2) Determine the effects of a fractionated exposure on sustained attention and recognition memory in comparison to the effects of single ion exposures on these measures; 3) Determine the effects of circadian disruptions and sleep fragmentation on sustained attention and recognition memory following radiation exposure (This aim has been modified in order to integrate with Nelson project); 4) Examine the underlying mechanisms of these deficits using immunohistochemical and pharmacogenetic procedures. ORIGINAL PROPOSAL DESCRIPTION: Assessing the biological consequences of living in the space radiation environment represents one of the highest priority areas of NASA research. Of critical importance is the need for an assessment of the vulnerabilities of the central nervous system (CNS) leading to functional neurobehavioral changes during long-term space missions, and the development of effective countermeasures to such risks. The present proposal addresses this need via the application of an animal model to 1) determine the effects of acute, single 16O and 4He ion exposures on sustained attention, social odor recognition memory, and social dominance; 2) determine the effects of a fractionated exposure on sustained attention and recognition memory in comparison to the effects of single ion exposures on these measures; 3) determine the effects of circadian disruptions and sleep fragmentation on sustained attention and recognition memory following radiation exposure; and 4) examine the underlying mechanisms of these deficits using immunohistochemical and pharmacogenetic procedures. Prior research has 1) identified rats that are sensitive to radiation-induced deficits in sustained attention and 2) shown that acute, single ion exposures alter social motivation and social odor recognition memory. The current proposal will determine how the immediate effects of irradiation impact subsequent neurobehavioral deficits by assessing various behavioral, physiological, and neurobiological markers of radiation exposure at early and later time points post-exposure. Groups of animals will be trained on a rodent version of the human psychomotor vigilance test, exposed to radiation, and then tested for both social processing and sustained attention deficits following exposure; subsets of rats will be implanted with biotelemetry devices to measure fluctuations in spontaneous locomotor activity and body temperatures following radiation exposure and changes in sleep or circadian disruptions. Individual variations in these behavioral, physiological, or neurobiological responses following radiation will be assessed. Likely mechanisms of damage to the CNS following radiation exposure will be examined using brain tissue, in addition to a pharmacogenetic technique to determine the severity of radiation-induced neurobehavioral deficits.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of the current project will be used to understand how radiation exposure affects the central nervous system to induce deficits in neurobehavioral function.
	During this grant year, we completed the behavioral data analysis for our study examining the effects of acute 4He (250 MeV/n) on neurobehavioral function. Acute 4He (250 MeV/n) study: This study examined the effects of an acute exposure to 5 or 25 cGy 4He (250 MeV/n) on social recognition and sustained attention, using the social odor recognition memory (SORM) test and the rPVT. The SORM was assessed at 4 time points following radiation exposure: 7 days, 30 days, 3 months, and 6 months. The rPVT was assessed between the 30 day time point and 6 month time point. At each time point, a subset of rats was sacrificed and brain tissue and blood plasma were collected for subsequent analysis. At 7 days following radiation exposure, only the 25 cGy-exposed rats displayed a deficit in social recognition. This group of rats also displayed a significant increase in plasma TNF-alpha levels compared to sham-irradiated control rats. Further, this group also showed significant decreases in specific types of bacteria in the GI microbiome (assessed via

	fecal samples), but no changes in GI architecture were noted. Increases in gut serotonin were also found in this group and suggest a potential mechanism connecting changes in the GI microbiome to deficits in social recognition memory at this early post-radiation time point. No changes were found in measures of the GI microbiome, serotonin, or social odor recognition memory in the 5 cGy-exposed subjects. A manuscript detailing these results was submitted for publication. At 30 days following radiation exposure, our preliminary analysis shows that both 5 and 25 cGy exposed rats displayed a deficit in social recognition, but no significant changes in plasma cytokine levels were apparent. Similar behavioral results were found for the 3 and 6 month time points, with both 4He exposed groups displaying deficits in social recognition. At 3 months, the 25 cGy rats displayed decreased plasma IL-6 levels. At 6 months, both IL-1beta and TNF-alpha were significantly decreased in the 25 cGy rats compared to the sham controls. A subset of 4He exposed rats (both 5 and 25 cGy) were identified as having a deficit in sustained attention, assessed with the rPVT. Our preliminary analysis shows that this subset of rats displayed significantly increased Q90 reaction times compared to their own pre-irradiation baselines and to the sham controls. Q90 reaction times are the slowest 10% of reaction times on the rPVT. These results suggest that radiation exposure in this subset of rats significantly slowed the slowest reaction times, which means that these animals had an even slower response to the stimulus than they did prior to radiation.
Task Progress:	4He-induced changes in bone: Through a tissue-sharing project with Anthony Lau, PhD at The College of New Jersey, we recently reported that 25 cGy 4He exposure results in significant decreases in the instantaneous modulus and relaxed shear modulus compared to both the sham control and 5 cGy-exposed rats. However, these changes recovered at the 6 month time point and this recovery was attributed to biological mechanisms involved in bone formation that were affected by radiation, bone turnover, and systemic change in hormones due to radiation exposure. This work was recently published in Calcified Tissue International (Thomas et al., 2021; see Bibliography section). GCR Dose-rate study: During this grant year, we completed behavioral analyses for the first half of our GCR dose-rate study. For this study, male (N=50) and female (N=50) rats were trained on the rPVT and shipped to BNL (Brookhaven National Laboratory) for exposure to the five-ion simplified GCR sim (50 cGy total dose). The simplified five-ion GCR sim consists of protons (250 and 1000 MeV), 4He (250 MeV/n), 16O (250 MeV/n), 28Si (600 MeV/n), 56Fe (600 MeV/n). The protracted exposure was delivered once per day (2.08 cGy), 6 days per week (12.5 cGy) for a total of 4 weeks. The acute exposure was delivered in one exposure at the end of the protracted exposure, such that all animals were approximately the same age when they had accumulated their total dose. At 7 days following each exposure, rats were tested on the SORM. Our preliminary analysis shows that all rats, regardless of sex or exposure, displayed deficits on the SORM. No sex differences have emerged at this time point. At 30 days following exposure, sex differences emerged, with only the female rats displaying deficits in social odor recognition memory. Both males groups displayed sham-like performances. At the 90 day time point, deficits were sex and dose-rate specific, with males exposed to the protracted GCRsim and females exposed to the acute GCRsim showing deficits. COVID impacts: Due to the impact of the COVID-19 pandemic, exposures of additional groups of rats were canceled and will be rescheduled in the upcoming years of the grant. Further, lab analyses of brain and blood samples were delayed and will be completed in the future. [Ed. note Jan 2022: Project continues with the same PI, Catherine Davis, with grant number 80NSSC22K0022, due to PI move to Henry M. Jackson Foundation.]

Bibliography Type:	Description: (Last Updated: 06/30/2026)
Articles in Peer-reviewed Journals	Thomas PK, Sullivan LK, Dickinson GH, Davis CM, Lau AG. "The effect of helium ion radiation on the material properties of bone." Calcif Tissue Int. 2021 Jun;108(6):808-818. https://doi.org/10.1007/s00223-021-00806-7 ; PMID: 33517470 , Jun-2021
Articles in Peer-reviewed Journals	Davis CM, Allen AR, Bowles DE. "Consequences of space radiation on the brain and cardiovascular system." J Environ Sci Health C Toxicol Carcinog. 2021;39(2):180-218. Published online: 27 Apr 2021. https://doi.org/10.1080/26896583.2021.1891825 ; PMID: 33902387 , Apr-2021
Articles in Peer-reviewed Journals	Sridharan V, Seawright JW, Landes RD, Cao M, Singh P, Davis CM, Mao XW, Singh SP, Zhang X, Nelson GA, Boerma M. "Effects of single-dose protons or oxygen ions on function and structure of the cardiovascular system in male Long Evans rats." Life Sci Space Res. 2020 Aug;26:62-8. https://doi.org/10.1016/j.lssr.2020.04.002 ; PMID: 32718688 , Aug-2020